

NAME:Atharv Revat Shukla

PRN:25070122053

Batch:2025-2029

ASSIGNMENT No: 12

TITLE: Write a Java program to create and use user-defined packages.

Problem Statement: Write a Java program to create and use user-defined packages.

Learning Objectives :To create and use user-defined packages for organizing Java programs.

Theory : Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Largescale software systems extensively use packages.

Exercise

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

```
package student;
```

```
public class Student {  
    int rollNo;  
    String name;
```

```
public Student(int rollNo, String name) {  
    this.rollNo = rollNo;  
    this.name = name;  
}  
  
public void display() {  
    System.out.println("Student Details");  
    System.out.println("Roll No : " + rollNo);  
    System.out.println("Name   : " + name);  
}  
}  
  
package faculty;  
  
public class Faculty {  
    int id;  
    String name;  
  
    public Faculty(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
  
    public void display() {  
        System.out.println("\nFaculty Details");  
        System.out.println("ID   : " + id);  
        System.out.println("Name : " + name);  
    }  
}  
  
import student.Student;
```

```
import faculty.Faculty;

public class Main {

    public static void main(String[] args) {

        Student s = new Student(101, "Rahul");
        Faculty f = new Faculty(201, "Dr. Sharma");

        s.display();
        f.display();
    }
}
```

2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
package library;

public class Book {

    int id;
    String title;
    String author;
    double price;

    public Book(int id, String title, String author, double price) {
        this.id = id;
        this.title = title;
        this.author = author;
        this.price = price;
    }
}
```

```
}
```

```
public void display() {  
    System.out.println("Book Details");  
    System.out.println("Book ID : " + id);  
    System.out.println("Title  : " + title);  
    System.out.println("Author : " + author);  
    System.out.println("Price  : " + price);  
}
```

```
}import library.Book;
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Book b = new Book(1, "Java Programming", "James Gosling", 599.50);  
  
        b.display();  
    }  
}
```